

CyberAudit

Database Migrations

7 migration files. 5 apps. 7 DB tables. 2 data seed migrations.

Project	CyberAudit Platform
Django version	5.1.15
Migration files	7 total (5 schema + 2 data seeds)
Apps	accounts / audits / reports / training / notifications
DB tables created	users / audit_packs / audit_requests / audit_reports / notifications / training_modules / training_progress
Data seeds	4 commercial packs (audits/0002) + 5 training modules (training/0002)
Seed strategy	RunPython + update_or_create, idempotent, re- runnable
Reversibility	All RunPython operations have a reverse function (remove_* deletes seeded rows)
Date	June 2026 / Juin 2026

1. Migration Tree & Execution Order

1. Arbre des migrations et ordre d'exécution

Django runs migrations in dependency order. The tree below shows the exact sequence and cross-app dependencies. reports and notifications both depend on audits/0002_seed_packs, ensuring packs exist before any report or notification references an AuditRequest.

Django exécute les migrations dans l'ordre des dépendances. L'arbre ci-dessous montre la séquence exacte et les dépendances inter-apps. Reports et notifications dépendent tous les deux de audits/0002_seed_packs, garantissant que les packs existent avant toute référence à un AuditRequest.

#	File	App / DB table	Type	Depends on
1	accounts/0001_initial	users	Schema	auth/0012
2	audits/0001_initial	audit_packs + audit_requests	Schema	accounts/0001
3	audits/0002_seed_packs	audit_packs (data)	Data seed	audits/0001
4	reports/0001_initial	audit_reports	Schema	audits/0002_seed_packs
5	notifications/0001_initial	notifications	Schema	audits/0002_seed_packs
6	training/0001_initial	training_modules + training_progress	Schema	accounts/0001
7	training/0002_seed_modules	training_modules (data)	Data seed	training/0001

Dependency graph

```
auth/0012_alter_user_first_name_max_length
├─ accounts/0001_initial (users)
│   └─ audits/0001_initial (audit_packs, audit_requests)
│       └─ audits/0002_seed_packs [DATA: 4 packs]
│           ├── reports/0001_initial (audit_reports)
│           └─ notifications/0001_initial (notifications)
└─ training/0001_initial (training_modules, training_progress)
    └─ training/0002_seed_modules [DATA: 5 modules]
```

2. accounts/0001_initial : users table

2. accounts/0001_initial : table users

Custom User model replaces Django's built-in auth.User. UUID primary key prevents sequential ID enumeration. Email is the login identifier (no username). Role field gates admin vs client access throughout the API.

Le modele User personnalise remplace auth.User de Django. La cle primaire UUID empêche l'enumeration sequentielle des IDs. L'email est l'identifiant de connexion (pas de username). Le champ role controle les acces admin vs client dans toute l'API.

```
class Migration(migrations.Migration):
    initial = True
    dependencies = [('auth', '0012_alter_user_first_name_max_length')]

    operations = [
        migrations.CreateModel(
            name='User',
            fields=[
                # Django auth fields (inherited)
                ('password', models.CharField(max_length=128)),
                ('last_login', models.DateTimeField(blank=True, null=True)),
                ('is_superuser', models.BooleanField(default=False)),

                # Custom fields
                ('id', models.UUIDField(primary_key=True,
default=uuid.uuid4, editable=False)),
                ('email', models.EmailField(max_length=254, unique=True)),
                ('first_name', models.CharField(max_length=50)),
                ('last_name', models.CharField(max_length=50)),
                ('company_name', models.CharField(max_length=100, blank=True)),
                ('role', models.CharField(max_length=10,

choices=[('client', 'Client'), ('admin', 'Administrateur')],
                default='client')),
                ('is_active', models.BooleanField(default=True)),
                ('is_staff', models.BooleanField(default=False)),
                ('created_at', models.DateTimeField(auto_now_add=True)),

                # Django permission M2M (kept for Django admin compatibility)
                ('groups', models.ManyToManyField(blank=True,
to='auth.group')),
                ('user_permissions', models.ManyToManyField(blank=True,
to='auth.permission')),
            ],
            options={'db_table': 'users', 'ordering': ['-created_at']},
        ),
    ]
```

UUID primary key

Prevents sequential ID guessing. Safe to expose in API URLs.
Empêche la devinette séquentielle des ID. Sécurise pour les URLs API.

email unique=True	Email is the USERNAME_FIELD. AUTH_USER_MODEL = 'accounts.User'. Email est USERNAME_FIELD. AUTH_USER_MODEL = 'accounts.User'.
role choices	'client' / 'admin'. Default: 'client'. All new sign-ups are clients. 'client' / 'admin'. Defaut 'client'. Toute inscription est client.
groups + user_permissions	Retained for Django Admin and per-object permission compatibility. Conserves pour Django Admin et la compatibilité permissions objet.
db_table = 'users'	Explicit table name. No app-prefix clutter (not 'accounts_user'). Nom de table explicite. Pas de préfixe app (pas 'accounts_user').

3. audits/0001_initial AuditPack + AuditRequest

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Two models in one migration. AuditPack defines the commercial catalogue (4 entries seeded in 0002). AuditRequest is the core business object: UUID PK, auto-generated reference, status state machine, FKs to client (PROTECT) and assigned_to (SET_NULL).

Deux modeles en une migration. AuditPack définit le catalogue commercial (4 entrees inserees en 0002). AuditRequest est l'objet metier central : UUID PK, référence auto-generee, machine d'etat status, FKs vers client (PROTECT) et assigned_to (SET_NULL).

AuditPack

```
migrations.CreateModel(
    name='AuditPack',
    fields=[
        ('id', models.BigAutoField(primary_key=True)), # auto int
        ('code', models.CharField(max_length=20, unique=True,
            choices=[('audit','Audit'),('security','Security'),
            ('protection','Protection'),('premium','Premium')])),
        ('name', models.CharField(max_length=50)),
        ('description', models.TextField(blank=True)),
        ('included_services', models.TextField(blank=True)),
        ('for_whom', models.TextField(blank=True)),
        ('perimeter', models.TextField(blank=True)),
        ('duration_days', models.PositiveSmallIntegerField()),
        ('price', models.DecimalField(max_digits=8, decimal_places=2)),
        ('created_at', models.DateTimeField(auto_now_add=True)),
        ('updated_at', models.DateTimeField(auto_now=True)),
    ],
    options={'db_table': 'audit_packs', 'ordering': ['price']},
)
```

AuditRequest

```
migrations.CreateModel(
    name='AuditRequest',
    fields=[
        ('id', models.UUIDField(primary_key=True, default=uuid.uuid4,
            editable=False)),
        ('reference', models.CharField(max_length=20, blank=True, unique=True,
            db_index=True)),
        ('status', models.CharField(max_length=20, db_index=True,
            default='pending',
            choices=[('pending','Pending'),('in_progress','In
            Progress'),('completed','Completed'),('archived','Archived')])),
        ('scope_notes', models.TextField(blank=True)),
        ('internal_notes', models.TextField(blank=True)),
        ('submitted_at', models.DateTimeField(auto_now_add=True)),
        ('updated_at', models.DateTimeField(auto_now=True)),
        ('completed_at', models.DateTimeField(blank=True, null=True)),
    ],
    options={'db_table': 'audit_requests', 'ordering': ['submitted_at']},
)
```

```

        # FKs
        ('client', models.ForeignKey(to=AUTH_USER_MODEL, on_delete=PROTECT,
related_name='audit_requests')),
        ('pack', models.ForeignKey(to='audits.AuditPack',
on_delete=PROTECT, related_name='requests')),
        ('assigned_to', models.ForeignKey(to=AUTH_USER_MODEL, on_delete=SET_NULL,
null=True, blank=True,
related_name='assigned_requests')),
    ],
    options={
        'db_table': 'audit_requests', 'ordering': ['-submitted_at'],
        'indexes': [models.Index(fields=['client', '-submitted_at'],
name='audit_reque_client__7a0a11_idx')]
    },
)

```

reference unique+blank	Blank on creation: populated by a pre_save signal with DOSSIER-YYYY-NNNN format. Vide a la création : alimente par un signal pre_save au format DOSSIER-YYYY-NNNN.
status db_index=True	Frequent filter target (admin dashboard). Index speeds SELECT WHERE status=... Cible de filtre frequente. L'index accélère les SELECT WHERE status=...
client PROTECT	Prevents user deletion while they have open requests. Empeche la suppression d'un utilisateur avec des demandes en cours.
assigned_to SET_NULL	Admin reassignment leaves request intact; assigned_to becomes NULL. Réaffectation admin laisse la demande intacte; assigned_to passe a NULL.
Index client + submitted_at	Client's own requests sorted by date, common dashboard query pattern. Demandes du client par date, pattern frequent sur le tableau de bord.

4. audits/0002_seed_packs : Data Migration

4. audits/0002_seed_packs -- Migration de données

Data migrations populate the audit_packs table at migrate-time. Using update_or_create makes the forward function idempotent: re-running never creates duplicates. The reverse function deletes only seeded rows (filter by known codes), leaving any user-created packs untouched.

Les migrations de données peuplent audit_packs au moment du migrate. L'utilisation de update_or_create rend la fonction forward idempotente : une re-exécution ne crée jamais de doublons. La fonction reverse supprime uniquement les lignes insérées (filtre par codes connus), laissant intacts les packs créés par l'utilisateur.

```
"""
0002_seed_packs.py -- Inserts 4 commercial packs (data migration).
"""
from django.db import migrations

PACKS = [
    {'code': 'audit',      'name': 'Pack Audit',      'price': 1000,
     'duration_days': 5,
     'for_whom': 'SMEs new to cybersecurity'},
    {'code': 'security',   'name': 'Pack Security',   'price': 2000,
     'duration_days': 10,
     'for_whom': 'SMEs that have experienced an incident'},
    {'code': 'protection', 'name': 'Pack Protection', 'price': 3500,
     'duration_days': 15,
     'for_whom': 'SMEs with sensitive data'},
    {'code': 'premium',    'name': 'Pack Premium',    'price': 5000,
     'duration_days': 20,
     'for_whom': 'SMEs seeking comprehensive coverage'},
]

def seed_packs(apps, schema_editor):
    AuditPack = apps.get_model('audits', 'AuditPack')
    for pack in PACKS:
        AuditPack.objects.update_or_create(code=pack['code'], defaults=pack)

def remove_packs(apps, schema_editor):
    AuditPack = apps.get_model('audits', 'AuditPack')
    AuditPack.objects.filter(code__in=[p['code'] for p in PACKS]).delete()

class Migration(migrations.Migration):
    dependencies = [('audits', '0001_initial')]
    operations = [migrations.RunPython(seed_packs, remove_packs)]
```

4 commercial packs

Code	Name	For whom	Price	Duration / Scope
audit	Pack Audit	SMEs new to cybersecurity	1 000 EUR	5 days: EBIOS RM risk analysis + vulnerability report. 5 jours: Analyse risques EBIOS RM + rapport.
security	Pack Security	Post-incident SMEs	2 000 EUR	10 days: Audit + incident management + remediation. 10 jours: Audit + gestion incidente + remédiation.
protection	Pack Protection	SMEs with sensitive data	3 500 EUR	15 days: Audit + SOC 24/7 + real-time monitoring. 15 jours: Audit + SOC 24/7 + surveillance temps réel.
premium	Pack Premium	Full coverage seekers	5 000 EUR	20 days: Audit + incidents + SOC + team awareness. 20 jours: Solution tout-en-un complète.

apps.get_model()	Uses the historical model (frozen at migration time), not the live model class. Required in data migrations to avoid issues when the model changes later. Utilise le modèle historique (gelé au moment de la migration). Obligatoire pour éviter les problèmes si le modèle change ensuite.
update_or_create	Idempotent: safe to run migrate multiple times or in CI. code is the lookup key. Idempotent : sûr de lancer migrate plusieurs fois ou en CI. code est la cle de recherche.
reverse remove_packs	Only deletes rows whose code is in the known list. User-added packs are safe. Supprime uniquement les lignes dont le code est dans la liste connue.
No model import	Data migrations never import from apps (e.g., from apps.audits.models import AuditPack). Always use apps.get_model(). Jamais d'import depuis apps. Toujours apps.get_model().

5. reports/0001 + notifications/0001

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Both migrations depend on audits/0002_seed_packs, not just 0001_initial. This ensures the 4 packs exist before any AuditRequest (and therefore any report or notification) can reference one. AuditReport uses OneToOneField (max 1 report per request). Notification UUID PK with composite index on (user, -created_at).

Les deux migrations dépendent de audits/0002_seed_packs, pas seulement 0001_initial. Cela garantit que les 4 packs existent avant tout AuditRequest (et donc tout rapport ou notification). AuditReport utilise OneToOneField (1 rapport max par demande). Notification a une PK UUID avec un index composite sur (user, -created_at).

reports/0001_initial : audit_reports

```
class Migration(migrations.Migration):
    dependencies = [('audits', '0002_seed_packs')] # ensures packs exist

    operations = [
        migrations.CreateModel(
            name='AuditReport',
            fields=[
                ('id', models.UUIDField(primary_key=True,
default=uuid.uuid4, editable=False)),
                ('audit_request', models.OneToOneField(to='audits.AuditRequest',
on_delete=CASCADE, related_name='report')),
                ('summary', models.TextField(blank=True)),
                ('verdict', models.CharField(max_length=255, blank=True)),
                ('grade', models.CharField(max_length=2, default='C',
choices=[('A', 'A -- Excellent'), ('B+', 'B+ --
Tres bon'),
                                ('B', 'B -- Bon'), ('C', 'C -- Moyen'),
                                ('D', 'D -- Faible'), ('E', 'E --
Mauvais'), ('F', 'F -- Critique')])),
                ('security_score', models.PositiveSmallIntegerField(default=50)),
                ('findings', models.JSONField(default=list, blank=True)),
                ('pdf_path', models.CharField(max_length=255, blank=True)),
                ('generated_at', models.DateTimeField(blank=True, null=True)),
                ('created_at', models.DateTimeField(auto_now_add=True)),
                ('updated_at', models.DateTimeField(auto_now=True)),
            ],
            options={'db_table': 'audit_reports', 'ordering': ['-created_at']},
        ),
    ]
```

notifications/0001_initial : notifications

```
class Migration(migrations.Migration):
    dependencies = [
        ('audits', '0002_seed_packs'),
        migrations.swappable_dependency(settings.AUTH_USER_MODEL),
    ]

    operations = [
        migrations.CreateModel(
            name='Notification',
            fields=[
                ('id', models.UUIDField(primary_key=True,
default=uuid.uuid4, editable=False)),
                ('type', models.CharField(max_length=32,
choices=[('request_received', 'Request
received'),
                ('status_changed', 'Status changed'),
                ('report_ready', 'Report ready'),
                ('generation_failed', 'Generation
failed')])),
                ('subject', models.CharField(max_length=255)),
                ('message', models.TextField()),
                ('status', models.CharField(max_length=10, db_index=True,
default='queued',
choices=[('queued', 'Queued'), ('sent', 'Sent'), ('failed', 'Failed')])),
                ('sent_at', models.DateTimeField(blank=True, null=True)),
                ('error_message', models.TextField(blank=True)),
                ('created_at', models.DateTimeField(auto_now_add=True)),
                ('user', models.ForeignKey(AUTH_USER_MODEL,
on_delete=CASCADE)),
                ('request', models.ForeignKey('audits.AuditRequest',
on_delete=CASCADE,
null=True, blank=True)),
            ],
            options={
                'db_table': 'notifications', 'ordering': ['-created_at'],
                'indexes': [models.Index(fields=['user', '-created_at'],
name='notificatio_user_id_...')]
            },
        ),
    ]
```

OneToOneField on AuditReport	Enforces 1 report per request at DB level. Attempting a second INSERT raises IntegrityError. Impose 1 rapport par demande au niveau DB. Une 2e insertion leve IntegrityError.
grade default C	New reports start at C/50 until Celery generates the PDF and services.py calculates the real grade. Les nouveaux rapports demarrent a C/50 jusqu'a ce que Celery calcule le vrai grade.
Notification status db_index	Celery worker queries pending/failed notifications. Index speeds this lookup. Le worker Celery interroge les notifications en attente. L'index accelere.
swappable_dependency	Future-proof AUTH_USER_MODEL swap. Correct pattern for any app that FKs to the user model. Echange futur de AUTH_USER_MODEL. Pattern correct pour tout FK vers le modele user.

6. training/0001_initial : TrainingModule + TrainingProgress

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Two models in one migration. TrainingModule is the course catalogue (slugged, levelled). TrainingProgress tracks per-user per-module state with a UniqueConstraint enforcing one progress row per (user, module) pair, the foundation for the idempotent start/complete API.

Deux modèles en une migration. TrainingModule est le catalogue de cours (sluge, nivele).

TrainingProgress suit l'état par utilisateur et par module avec un UniqueConstraint imposant une seule ligne de progression par paire (user, module), base de l'API start/complete idempotente.

```
class Migration(migrations.Migration):
    initial = True
    dependencies = [migrations.swappable_dependency(settings.AUTH_USER_MODEL)]

    operations = [
        migrations.CreateModel(
            name='TrainingModule',
            fields=[
                ('id', models.BigAutoField(primary_key=True)),
                ('slug', models.SlugField(max_length=80, unique=True)),
                ('title', models.CharField(max_length=200)),
                ('description', models.TextField(blank=True)),
                ('content_md', models.TextField(blank=True)),
                ('duration_min', models.PositiveSmallIntegerField(default=10)),
                ('level', models.CharField(max_length=20,
                    default='beginner',
                    choices=[('beginner', 'Debutant'),
                           ('intermediate', 'Intermediaire'),
                           ('advanced', 'Avance')])),
                ('published_at', models.DateTimeField(blank=True, null=True)),
                ('created_at', models.DateTimeField(auto_now_add=True)),
                ('updated_at', models.DateTimeField(auto_now=True)),
            ],
            options={'db_table': 'training_modules', 'ordering': ['id']},
        ),
        migrations.CreateModel(
            name='TrainingProgress',
            fields=[
                ('id', models.BigAutoField(primary_key=True)),
                ('status', models.CharField(max_length=20, db_index=True,
                    default='to_start',
                    choices=[('to_start', 'A commencer'),
                           ('in_progress', 'En cours'),
                           ('completed', 'Termine')])),
                ('started_at', models.DateTimeField(blank=True, null=True)),
                ('completed_at', models.DateTimeField(blank=True, null=True)),
                ('created_at', models.DateTimeField(auto_now_add=True)),
                ('updated_at', models.DateTimeField(auto_now=True)),
                ('module', models.ForeignKey(to='training.TrainingModule',
                    on_delete=CASCADE)),
                ('user', models.ForeignKey(to=AUTH_USER_MODEL,
                    on_delete=CASCADE)),
            ],
            options={
```

```

        'db_table': 'training_progress', 'ordering': ['-updated_at'],
        'constraints': [
            models.UniqueConstraint(fields=('user', 'module'),
name='uniq_user_module')
        ],
    },
),
]

```

slug unique	Human-readable identifier for URLs and seed data. Immutable once published. Identifiant lisible pour les URLs et données de seed. Immuable une fois publié.
content_md TextField	Markdown content stored in DB. Frontend renders it via a Markdown renderer. Contenu Markdown stocké en DB. Le frontend le rend via un moteur Markdown.
UniqueConstraint (user, module)	DB-level guarantee: one row per user/module pair. Prevents duplicate progress records. Garantie DB : une ligne par paire user/module. Empêche les doublons.
status to_start default	Row is created with to_start on first access. No row = not started. Ligne créée avec to_start au premier accès. Pas de ligne = pas commence.
ordering = ['-updated_at']	Progress list ordered by most recently updated -- natural dashboard sort. Liste ordonnée par mise à jour récente -- tri naturel pour le tableau de bord.

7. training/0002_seed_modules : Data Migration

7. training/0002_seed_modules : Migration de donnees

Inserts 5 cybersecurity awareness modules at migrate-time. Each module has a slug (lookup key), title, Markdown content, duration, and level. `timezone.now()` stamps `published_at` so all seeded modules are immediately visible to users without an admin publish step.

Insérer 5 modules de sensibilisation a la cybersecurite au moment du migrate. Chaque module a un slug (cle de recherche), un titre, du contenu Markdown, une durée et un niveau. `timezone.now()` initialise `published_at` pour que tous les modules seeds soient immédiatement visibles sans étape de publication admin.

```
"""training/0002_seed_modules.py -- Pre-insertion of 5 base modules."""
from django.db import migrations
from django.utils import timezone

MODULES = [
    {'slug': 'anti-phishing', 'title': 'Anti-phishing',
     'duration_min': 10, 'level': 'beginner'},
    {'slug': 'mfa', 'title': 'Strong passwords & MFA',
     'duration_min': 8, 'level': 'beginner'},
    {'slug': 'wifi-vpn', 'title': 'Public Wi-Fi & VPN',
     'duration_min': 12, 'level': 'intermediate'},
    {'slug': 'data-backup', 'title': 'Data backup',
     'duration_min': 15, 'level': 'intermediate'},
    {'slug': 'incident-response', 'title': 'Incident response',
     'duration_min': 20, 'level': 'advanced'},
]

def seed_modules(apps, schema_editor):
    TrainingModule = apps.get_model('training', 'TrainingModule')
    now = timezone.now()
    for module in MODULES:
        TrainingModule.objects.update_or_create(
            slug=module['slug'],
            defaults=**module, 'published_at': now,
        )

def remove_modules(apps, schema_editor):
    TrainingModule = apps.get_model('training', 'TrainingModule')
    TrainingModule.objects.filter(slug__in=[m['slug'] for m in MODULES]).delete()

class Migration(migrations.Migration):
    dependencies = [('training', '0001_initial')]
    operations = [migrations.RunPython(seed_modules, remove_modules)]
```

5 seeded training modules

Slug	Title	Level	Duration	Key learning EN / FR
anti-phishing	Anti-phishing	Beginner	10 min	Spot warning signs: unknown sender, fake urgency, strange links. Reperer les signaux d'alerte.
mfa	Strong passwords & MFA	Beginner	8 min	12 chars minimum, password manager, TOTP everywhere. 12 chars min, gestionnaire, TOTP partout.
wifi-vpn	Public Wi-Fi & VPN	Intermediate	12 min	Avoid open hotspots, always VPN on public networks. Eviter les hotspots ouverts, toujours VPN.
data-backup	Data backup	Intermediate	15 min	3-2-1 rule, encrypted off-site backups. Règle 3-2-1, sauvegardes chiffrées hors site.
incident-response	Incident response	Advanced	20 min	Identify, contain, eradicate, restore, lessons learned. Identifier, contenir, éradiquer, restaurer.

8. Design & Security Summary

8. Recapitulatif conception et sécurité

Feature	Detail EN / FR
UUID PKs	accounts/audits.AuditRequest/reports/notifications use UUID v4 : no sequential enumeration in API URLs. PKs UUID v4 : pas d'énumération séquentielle dans les URLs API.
Dependency on 0002	reports + notifications depend on audits/0002_seed_packs, not 0001. Guarantees data integrity: no AuditRequest can exist without a valid pack. reports + notifications dependent de audits/0002_seed_packs. Intégrité garantie : pas d'AuditRequest sans pack valide.
PROTECT on FKs	client + pack use on_delete=PROTECT. Deleting a user or pack with linked requests is blocked at DB level. client + pack utilisent PROTECT. Suppression bloquée au niveau DB si des demandes existent.
OneToOneField	Enforces max 1 AuditReport per AuditRequest at DB level, no application-layer check needed. Impose max 1 rapport par demande au niveau DB, pas de vérification applicative nécessaire.
UniqueConstraint	training_progress (user, module) enforced by DB. The idempotent start/complete API relies on this. Contrainte DB sur (user, module). L'API start/complète idempotente repose sur ce mécanisme.
update_or_create seeds	Both data migrations use update_or_create, safe for CI, re-deployments, and fresh installs. Les deux migrations de données utilisent update_or_create, sur en CI, re-déploiements et nouvelles installations.
Reversible seeds	Every RunPython has a remove_* reverse. python manage.py migrate <app> zero works cleanly. Chaque RunPython a une fonction inverse remove_*. migrate <app> zéro fonctionne proprement.
apps.get_model()	Data migrations never import live models, uses historical model snapshot. Prevents future migration conflicts. Les migrations de données n'importent jamais les modèles en direct. Utilise le snapshot historique.
db_index on status	audit_requests.status and notifications.status are db_index=True. Dashboards filter by status constantly.

	audit_requests.status et notifications.status sont indexés. Les tableaux de bord filtrent en permanence par status.
Composite index	audit_requests: Index(client, -submitted_at). notifications: Index(user, -created_at). Covers the two most common list queries. Index(client, -submitted_at) et Index(user, -created_at) couvrent les deux requêtes de liste les plus fréquentes.